

PROJECT TITAN: 2-GIGAWATT AI DATA CENTER

Unified Investment Committee Dossier & Multi-Model Economic Synthesis

Location: Texas ERCOT Interconnect | **Power Rating:** 2,000 Megawatts | **Date:** October 2026 | **Classification:** Confidential

This master memorandum synthesizes the economic, civil, electrical, and thermal engineering evaluations conducted across **8 leading AI model architectures** for the development of a mega-scale **2.0 Gigawatt (GW) AI computing campus in Texas**. With the AI compute transition driving rack densities from 10 kW toward 100+ kW, traditional data center development models fail. This report establishes institutional CapEx benchmarks, examines substation interconnection constraints with the ERCOT transmission grid, details cooling requirements, and consolidates all model findings into an unified investment roadmap.

Table 1.0: Cross-Model Economic Valuation & Architecture Matrix

Model Name	CapEx Low	CapEx High	CapEx Mid	Annual OpEx	\$/Watt	Inference
Google Gemma 4 E4B	\$2,900M	\$5,600M	\$4,250M	\$1,600M	\$2.12/W	9.3 tps
Llama 3.3 70B Instruct	\$543M	\$943M	\$743M	\$120M	\$0.37/W	0.7 tps
Mistral Small 24B Instruct	\$2,422M	\$2,664M	\$2,664M	\$100M	\$1.33/W	1.4 tps
Cerberus 4B v2 Abliterated	\$1,800M	\$3,200M	\$2,500M	\$210M	\$1.25/W	56.5 tps
DeepSeek R1 Distill Qwen 14B	\$925M	\$1,000M	\$962M	\$100M	\$0.48/W	3.4 tps
Cerberus v0.1	\$500M	\$750M	\$625M	\$85M	\$0.31/W	0.8 tps
Qwen 2.5 32B Instruct	\$88M	\$1,750M	\$875M	\$40M	\$0.44/W	1.1 tps
Qwen 2.5 Coder 14B Instruct	\$90M	\$1,500M	\$750M	\$65M	\$0.38/W	3.2 tps

Key Synthesis Findings & Capital Allocation Themes

- 1. Electrical Infrastructure Dominance (45%–55% of CapEx):** The primary capital hurdle is not the building envelope, but high-voltage substations, dual-circuit 345kV ERCOT interconnects, and N+2 static UPS systems totaling \$1.8B–\$2.5B alone.
- 2. Thermal Rejection & Direct Liquid Cooling (DLC):** High ambient Texas summer conditions (>100°F) combined with 100 kW/rack GPUs render pure air cooling infeasible. Liquid-to-chip closed loop and immersion cooling are mandatory to protect PUE.
- 3. Realistic Institutional Valuation Range:** While basic powered-shell estimates floor around \$625M–\$1.0B, a fully commissioned turnkey AI campus ranges between **\$2.4B and \$5.6B**.

SECTION 2: INDIVIDUAL MODEL ARCHITECTURAL PROFILES

Google Gemma 4 E4B — Detailed Assessment

Strategic Focus: Institutional Investment Memorandum, ERCOT 345kV Grid Tie, High-Voltage Electrical Dominance

Synthesis: Provides the most realistic utility-scale valuation (\$4.25B mid). Explicitly identifies the ERCOT substation, dual-feeder redundancy, and liquid cooling infrastructure as representing over 60% of total CapEx, factoring in continuous 2GW transmission tariffs.

Cost Component	Allocation (\$USD)	% Share
Site Acquisition & ERCOT Interconnection Permitting	\$100,000,000	2.4%
Civil & Structural Mega-Buildout (Reinforced Foundations)	\$575,000,000	13.5%
Electrical Infrastructure (Dual 345kV Substations, UPS, Switchgear)	\$2,150,000,000	50.6%
Cooling & Direct Liquid Cooling (DLC) Plant (CDU / Closed Loop)	\$475,000,000	11.2%
Core High-Density Data Hall Buildout (Gas Suppression, Fiber)	\$350,000,000	8.2%
Initial Network Backbone & Commissioning	\$200,000,000	4.7%
Contingency & Supply Chain Escalation Buffer	\$400,000,000	9.4%
TOTAL MODELED CAPEX	\$4,250,000,000	100.0%

Llama 3.3 70B Instruct — Detailed Assessment

Strategic Focus: Granular MEP, Tier IV Physical Security Hardening, Commissioning Schedules

Synthesis: Excels in component-level civil, structural, and mechanical pricing. Models physical security (biometrics, vehicle barriers, perimeter fencing) and MEP distribution down to unit costs per square foot.

Cost Component	Allocation (\$USD)	% Share
Land & Site Preparation (50 Acres)	\$15,000,000	3.1%
Building & Structural Shell (Steel Cladding, Roofing)	\$66,000,000	13.8%
MEP (Power Distribution Units, Generators, Chillers)	\$122,500,000	25.6%
Data Center Infrastructure (Racks, PDU Runs)	\$75,000,000	15.7%
Security & Access Control (Biometric, CCTV, Perimeter)	\$18,500,000	3.9%
IT & Optical Network Infrastructure (Spine-Leaf Fiber)	\$24,500,000	5.1%
Testing, Factory Acceptance & Commissioning	\$7,500,000	1.6%
Contingency Buffer (10% Risk Allocation)	\$150,000,000	31.3%
TOTAL MODELED CAPEX	\$479,000,000	100.0%

Mistral Small 24B Instruct — Detailed Assessment

Strategic Focus: Power-Scaled Metric Breakdown (2,000,000 kW Unit Formula)

Synthesis: Grounds the economic architecture directly on the 2,000,000 kW electrical load, assigning rigorous \$/kW benchmarks across server equipment (\$500/kW), power gear (\$300/kW), and cooling plants (\$100/kW).

Cost Component	Allocation (\$USD)	% Share
Land Acquisition (50 Acres Greenfield)	\$10,000,000	0.4%
Building Shell (\$300/sq ft @ 1M sq ft)	\$300,000,000	11.3%
Data Center Compute Infrastructure (\$500/kW @ 2M kW)	\$1,000,000,000	37.5%
Power Infrastructure (\$300/kW @ 2M kW)	\$600,000,000	22.5%

Cooling Systems (\$100/kW @ 2M kW)	\$200,000,000	7.5%
Security & High-Speed Cabling	\$62,000,000	2.3%
AI Hardware Accelerators & Software Licensure	\$250,000,000	9.4%
Contingency Reserve (10% Overrun Buffer)	\$242,200,000	9.1%
TOTAL MODELED CAPEX	\$2,664,200,000	100.0%

Cerberus 4B v2 Abliterated — Detailed Assessment

Strategic Focus: Texas Regional Tradeoffs (DFW vs Permian Basin), Two-Phase Immersion Racks
Synthesis: Comprehensive geographic and cooling analysis comparing power tariffs and water scarcity across Dallas-Fort Worth and West Texas, stipulating dielectric immersion cooling to withstand high ambient temperatures.

Cost Component	Allocation (\$USD)	% Share
Site Acquisition & Strategic Water Rights (ERCOT)	\$80,000,000	3.2%
Reinforced Structural Hardening & Thermal Envelope	\$420,000,000	16.8%
Substation & Dual-Grid High Voltage Infeeds	\$1,100,000,000	44.0%
Two-Phase Liquid Immersion Cooling Plant	\$450,000,000	18.0%
Spine-Leaf Optical Compute Matrix	\$300,000,000	12.0%
Contingency Buffer & Project Management	\$150,000,000	6.0%
TOTAL MODELED CAPEX	\$2,500,000,000	100.0%

DeepSeek R1 Distill Qwen 14B — Detailed Assessment

Strategic Focus: Reasoning Analysis, Tier IV 99.995% Uptime, 5-Year Cumulative TCO
Synthesis: Combines a detailed reasoning trace with a balanced baseline for a 1M sq ft facility, incorporating environmental assessments, workforce development, and five-year operational reserve projections.

Cost Component	Allocation (\$USD)	% Share
Dallas-Fort Worth Land Acquisition (50 Acres)	\$15,000,000	1.1%
Tier IV Building Construction (\$350/sq ft)	\$350,000,000	25.5%
Power Distribution & Chiller Systems	\$150,000,000	10.9%
High-Performance Servers & Networking	\$250,000,000	18.2%
Permits, Environmental & Training Programs	\$40,000,000	2.9%
Contingency Reserve (10%)	\$70,000,000	5.1%
5-Year Cumulative Operating Reserve	\$500,000,000	36.4%
TOTAL MODELED CAPEX	\$1,375,000,000	100.0%

Cerberus v0.1 — Detailed Assessment

Strategic Focus: Base Greenfield Core & Shell Estimate (Powered Shell Model)
Synthesis: Models the bare core and powered shell tier, appropriate for a wholesale data center landlord where hyperscale tenants finance their own GPU clusters and proprietary cooling racks.

Cost Component	Allocation (\$USD)	% Share
Greenfield Site Land Acquisition	\$10,000,000	1.6%
Site Grading & Utilities Installation	\$5,000,000	0.8%
Shell Building Construction	\$30,000,000	4.8%

Core Mechanical & Electrical Systems	\$450,000,000	72.0%
Commissioning & Fit-Out	\$110,000,000	17.6%
Contingency Funds	\$20,000,000	3.2%
TOTAL MODELED CAPEX	\$625,000,000	100.0%

Qwen 2.5 32B Instruct — Detailed Assessment

Strategic Focus: Modular Phase 1 Pilot to 2GW Scale-Out (De-risked Rollout)
Synthesis: Provides a modular 100k sq ft Phase 1 proof-of-concept (\$87.5M) expandable across 20 modular data halls to fulfill the entire 2GW footprint without requiring day-one full capital drawdowns.

Cost Component	Allocation (\$USD)	% Share
Phase 1 Land Acquisition (50 Acres)	\$500,000	0.0%
Site Prep & Utilities	\$200,000	0.0%
Phase 1 Structural Buildout	\$30,000,000	1.7%
Electrical Wiring & Switchgear	\$10,000,000	0.6%
HVAC & Chiller Modules	\$10,000,000	0.6%
Data Center Infrastructure Servers	\$20,000,000	1.1%
Security & Access Control	\$1,000,000	0.1%
Phase 1 Contingency	\$5,000,000	0.3%
Full 2GW Scale Multiplier (Modular Expansion)	\$1,662,500,000	95.6%
TOTAL MODELED CAPEX	\$1,739,200,000	100.0%

Qwen 2.5 Coder 14B Instruct — Detailed Assessment

Strategic Focus: 8-Sheet Parameterized Financial Engine Schema
Synthesis: Outlines an engineering spreadsheet architecture spanning land acquisition, architectural fees, MEP, optical network fabrics, and construction debt amortization.

Cost Component	Allocation (\$USD)	% Share
Land & Civil Works	\$15,000,000	2.0%
Architectural, Structural & MEP Engineering	\$35,000,000	4.7%
High-Voltage Power Systems & Generators	\$350,000,000	46.7%
High-Density Chiller & Cooling Plants	\$180,000,000	24.0%
High-Speed Optical Fabric & Equipment	\$120,000,000	16.0%
Contingency & Financing Debt Reserve	\$50,000,000	6.7%
TOTAL MODELED CAPEX	\$750,000,000	100.0%

SECTION 3: ERCOT GRID RISK, SENSITIVITY & IC SIGN-OFF

Power Grid Curtailment & Pricing Risk: A 2.0 GW load running at 95% utilization consumes 16,644,000 MWh annually. In the Texas ERCOT market, summer peak demand spikes (4CP charges) can surge spot power from \$45/MWh to over \$5,000/MWh. To mitigate this risk, the investment model prescribes long-term indexed Power Purchase Agreements (PPAs) combined with a 500 MW on-site battery energy storage system (BESS) and dual-fuel backup generation.

Table 3.0: Annualized Power OpEx vs Tariff & PUE Efficiency

Tariff (\$/MWh)	Annual MWh	Base Power OpEx	PUE 1.15 (DLC)	PUE 1.25 (Hybrid)	PUE 1.40 (Air)
\$45.00/MWh	16,644,000	\$748,980,000	\$861,327,000	\$936,225,000	\$1,048,572,000
\$65.00/MWh	16,644,000	\$1,081,860,000	\$1,244,139,000	\$1,352,325,000	\$1,514,604,000
\$85.00/MWh	16,644,000	\$1,414,740,000	\$1,626,951,000	\$1,768,425,000	\$1,980,636,000
\$100.00/MWh	16,644,000	\$1,664,400,000	\$1,914,060,000	\$2,080,500,000	\$2,330,160,000

Investment Committee Recommendation & Execution Milestone

Recommendation: PROCEED to Phase 1 Interconnection Study (FIS) with ERCOT and secure 500-acre site options in the Oncor/CenterPoint service territory. Initial tranche deployment of **\$650M for Phase 1 (250 MW)** is authorized, contingent on binding hyperscale compute offtake agreements.

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Date: 2026-10-04	Date: 2026-10-04	Date: 2026-10-04
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